## Highlights

A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

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- A signal-to-drift ratio quantifies fault detectability without a model or detector
- Phasor residuals remove the faulted-phase dependence of negative-sequence ratios
- Signatures relocate across voltage, controller and field current by machine and drive
- A decision table selects the cheapest adequate sensor suite with bootstrap confidence
- Learned detectors transfer across generator types under healthy-referenced features